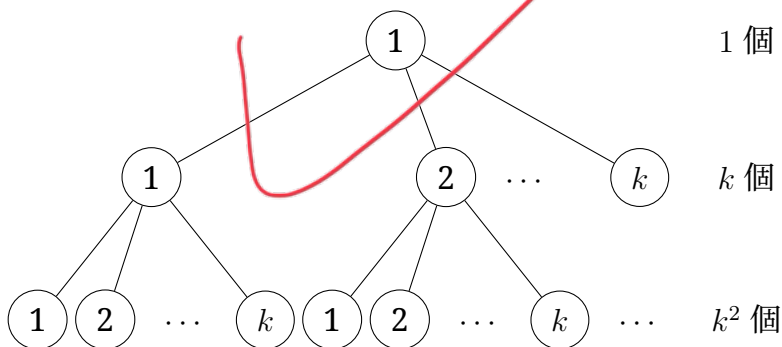


1. (10%) What is the maximum number of nodes in a k -ary tree of height h ? Prove your answer.

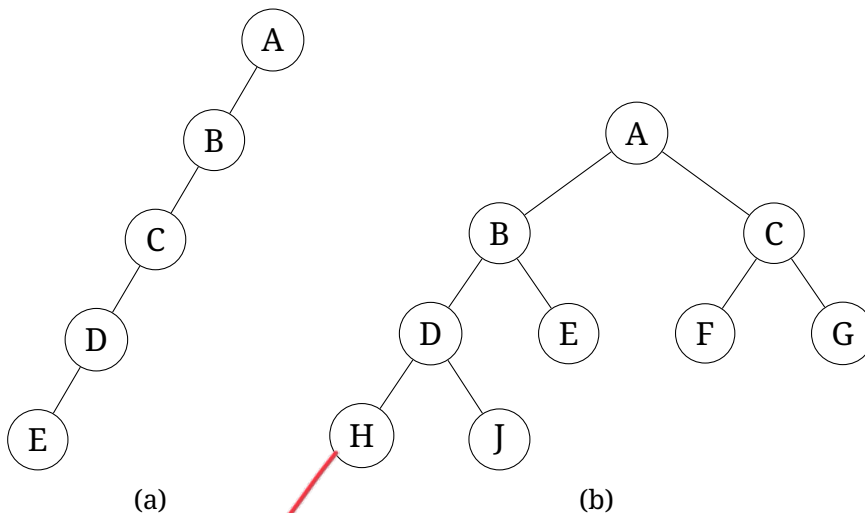
Ans: 每一個節點都會分支出 k 個節點，因此第 i 層有 k^i 個節點，故

$$\begin{aligned} \text{所求} &= \underbrace{1 + k + k^2 + \dots + k^{h-1}}_{\text{共 } h \text{ 層}} \\ &= \sum_{i=0}^{h-1} k^i \\ &= \frac{k^h - 1}{k - 1} \end{aligned}$$



□

2. (20%) Write out the inorder, preorder, postorder, and level-order travels for the binary trees (a) and (b).



Ans (a): Inorder: E D C B A
Preorder: A B C D E
Postorder: E D C B A
Level-order: A B C D E

□

Ans (b): Inorder: H D J B E A F C G
Preorder: A B D H J E C F G
Postorder: H J D E B F G C A
Level-order: A B C D E F G H J

□

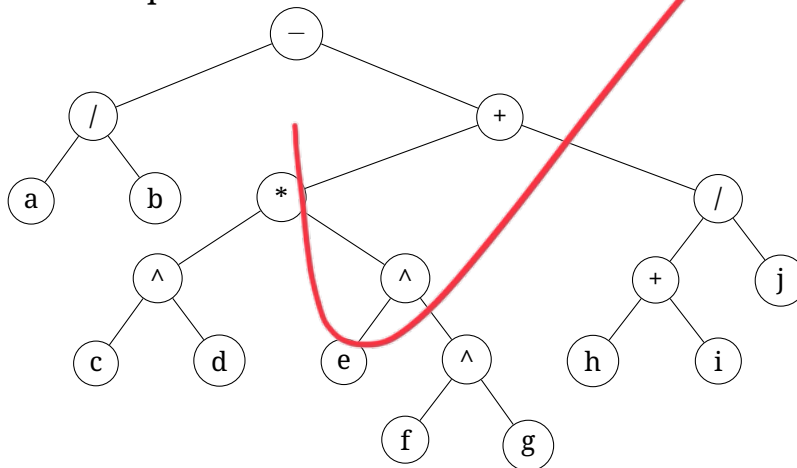
3. (15%) Considering operator precedence, draw the expression tree of the infix expression. Note that $\wedge$ is power.

$$a / b - c \wedge d * e \wedge f \wedge g + (h + i) / j$$

Ans:

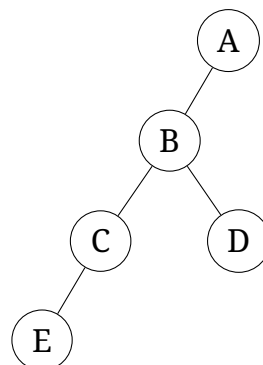
$$\left[\frac{a}{b} \right] - \left[(c \wedge d) * (e \wedge f \wedge g) \right] + \left[\frac{(h + i)}{j} \right]$$

所以其 expression tree 為

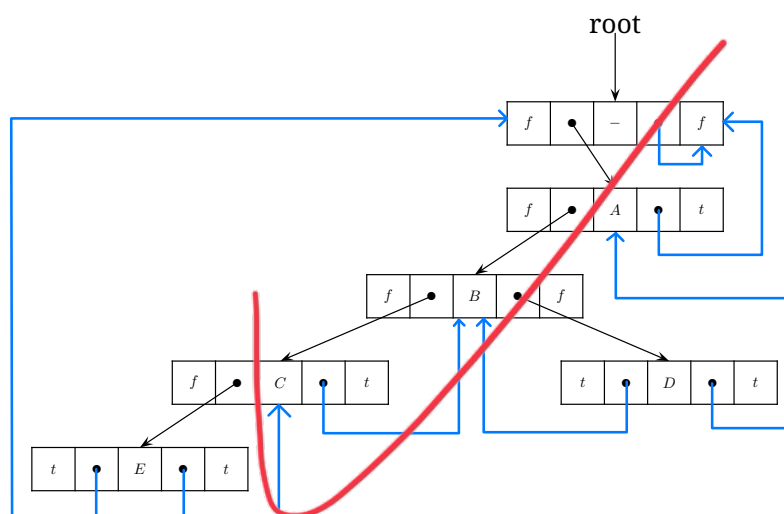


□

4. (10%) Draw the threaded representation of the following binary tree.



Ans: Threaded Tree:



□

5. (10%) Suppose that we execute the following instructions:

**push 7, push 9, push 16, push 30, push 49, pop, push 82, push 5, push 33,
push 31, push 6, pop, push 2, push 1.**

(a) What is the resultant heap if we use a max heap?

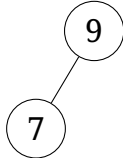
(b) What is the resultant heap if we use a min heap?

Ans (a): 使用最大堆積

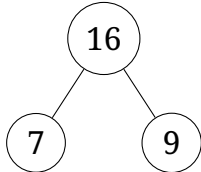
1. **push 7**



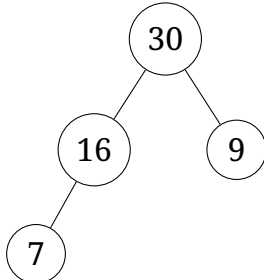
2. **push 9**



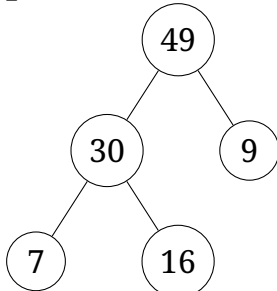
3. **push 16**



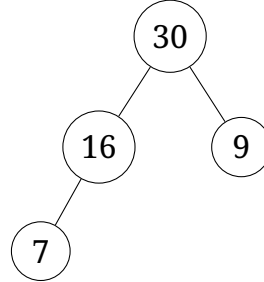
4. **push 30**



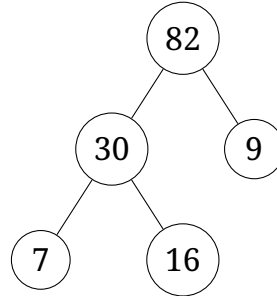
5. **push 49**



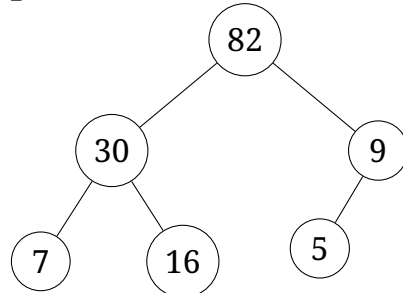
6. **pop**



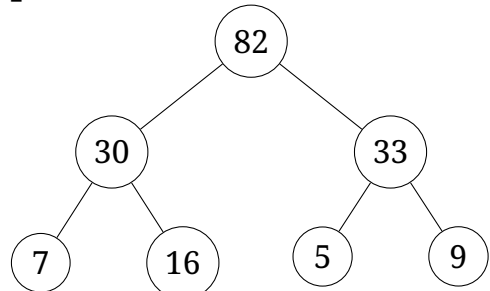
7. **push 82**



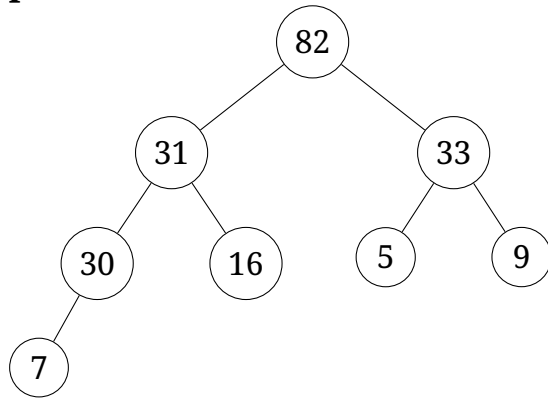
8. **push 5**



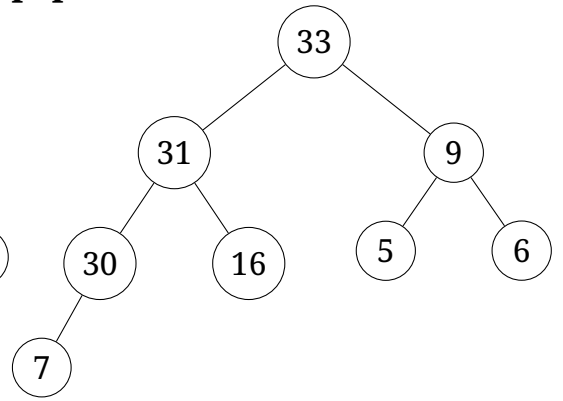
9. **push 33**



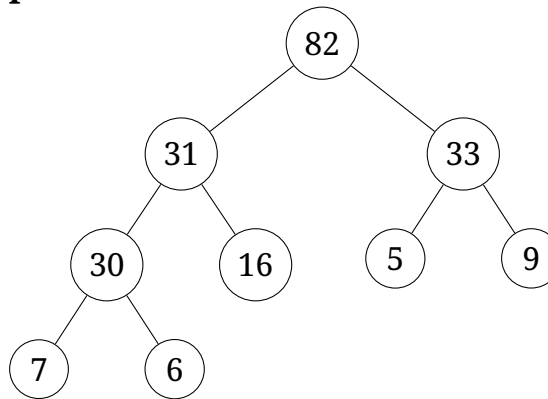
10. **push 31**



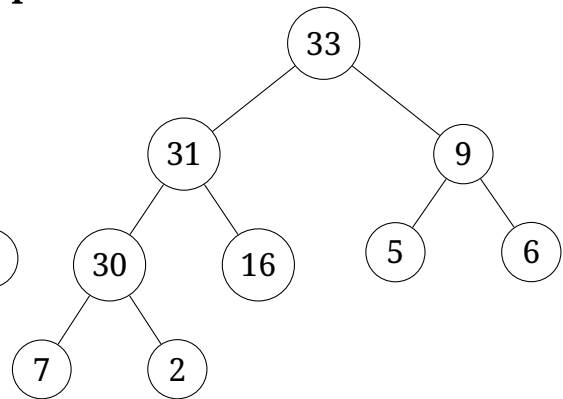
12. **pop**



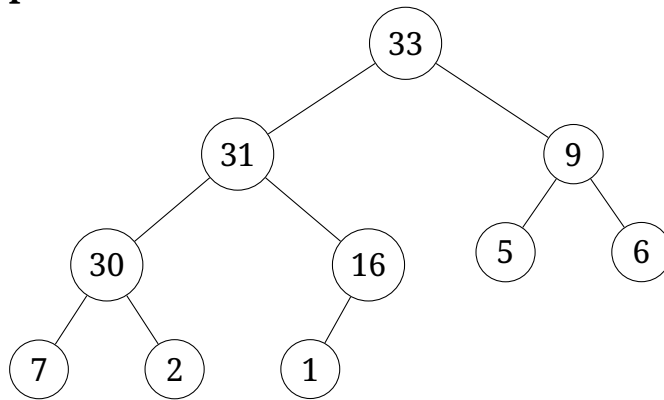
11. **push 6**



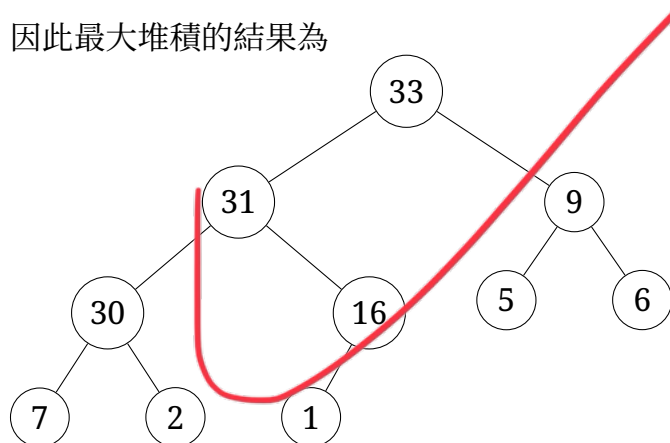
13. **push 2**



14. **push 1**



因此最大堆積的結果為



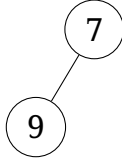
□

Ans (b): 使用最小堆積

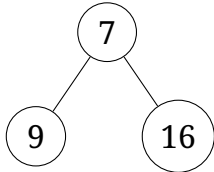
1. **push 7**



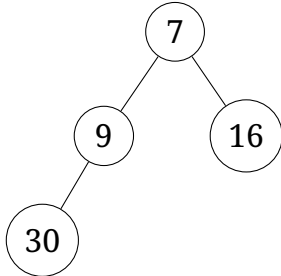
2. **push 9**



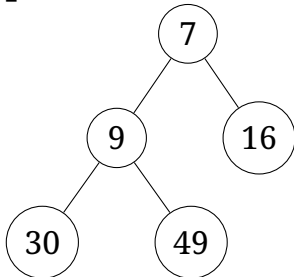
3. **push 16**



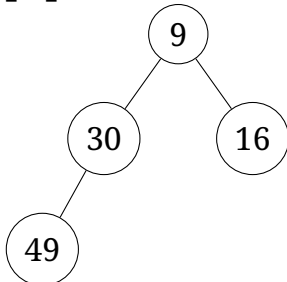
4. **push 30**



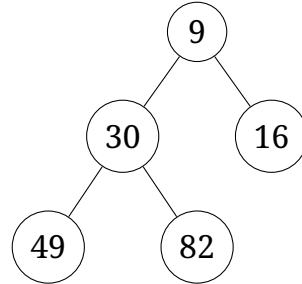
5. **push 49**



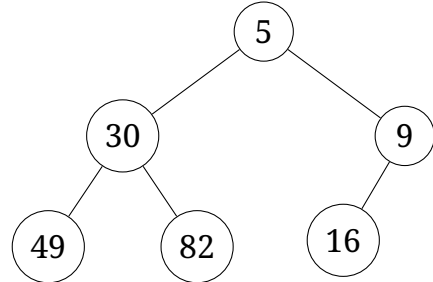
6. **pop**



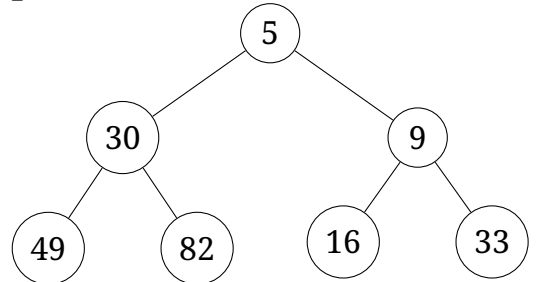
7. **push 82**



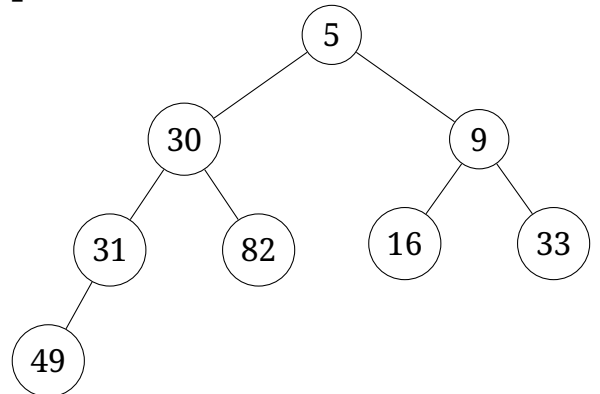
8. **push 5**



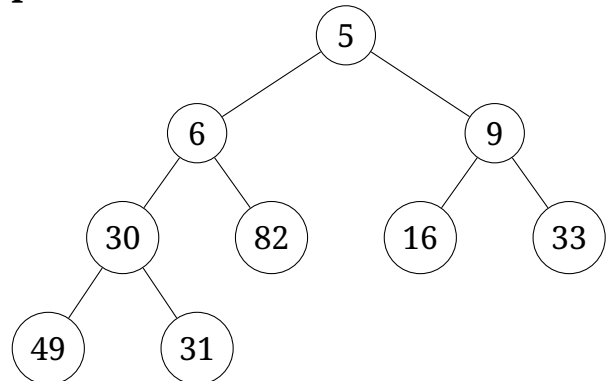
9. **push 33**



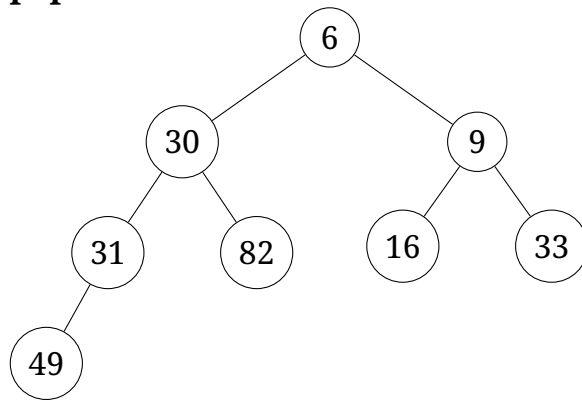
10. **push 31**



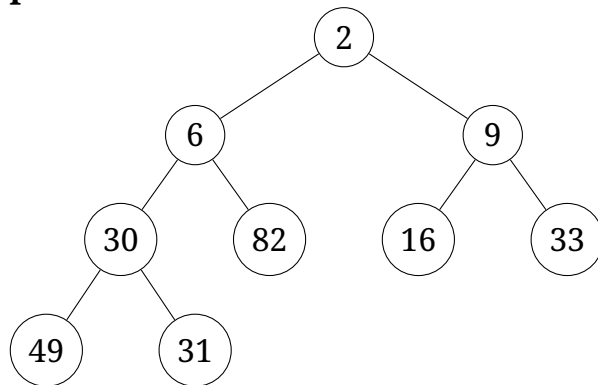
11. **push 6**



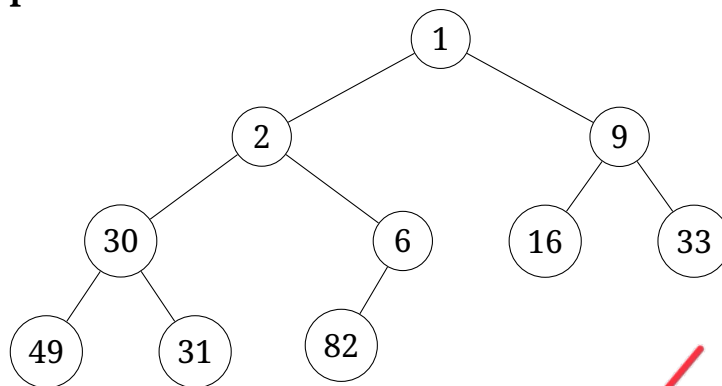
12. **pop**



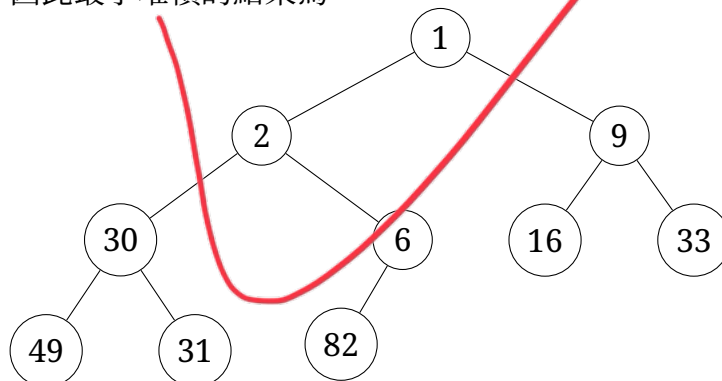
13. **push 2**



14. **push 1**



因此最小堆積的結果為



□

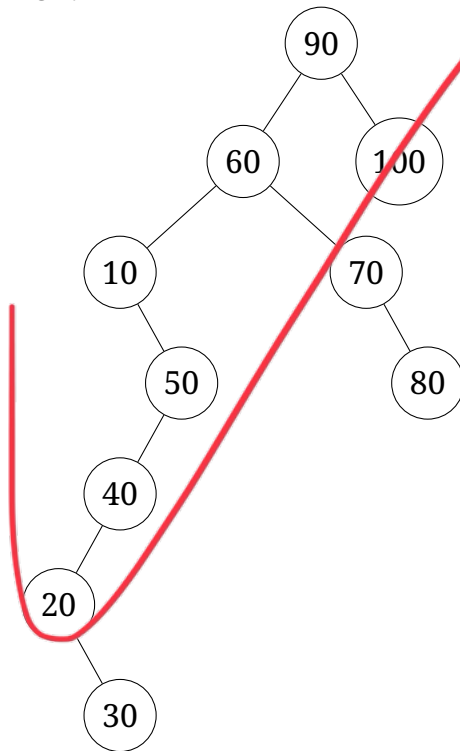
6. (10%) Given the following numbers:

90, 60, 70, 100, 10, 50, 40, 20, 30, 80.

(a) Construct a binary search tree.

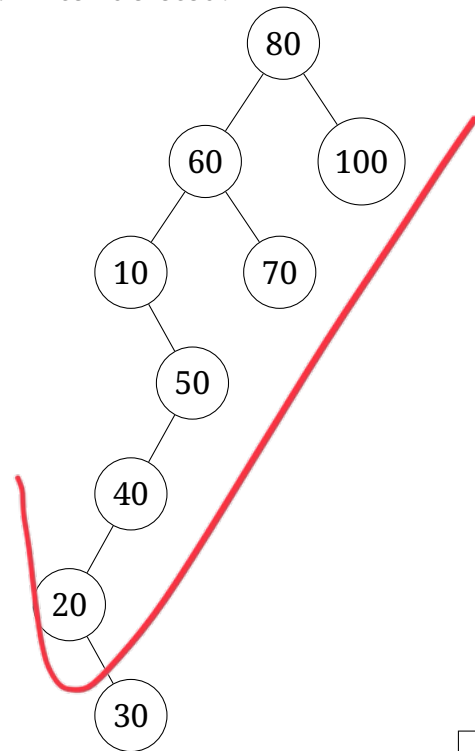
(b) From (a), what does the binary search tree become if we delete 90 using the rule “replaced by the largest element in its left subtree”?

Ans (a): BST:



□

Ans (b): After deleted:



□

7. (12%) How many distinct 6-node binary tree structures can we create?

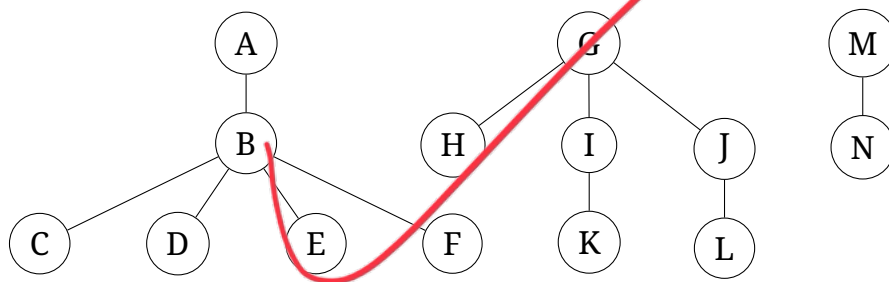
Ans:

$$\begin{aligned}
 B(6) &= \frac{1}{6+1} \binom{12}{6} \\
 &= \frac{1}{7} \times \frac{12 \times 11 \times 10 \times 9 \times 8 \times 7}{1 \times 2 \times 3 \times 4 \times 5 \times 6} \\
 &= 11 \times 3 \times 2^2 \\
 &= 132
 \end{aligned}$$

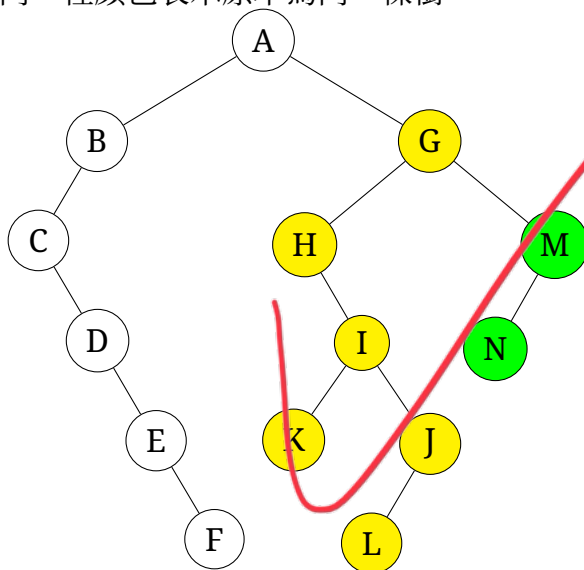
□

8. (13%) Transforming the forest into a binary tree using the rule:

- If $T_1, \dots, T_n$ is a forest of trees, then the binary tree corresponding to this forest, denoted by $B(T_1, \dots, T_n)$
- Root equal to $\text{root}(T_1)$
- Left subtree equal to $B(T_{11}, T_{12}, \dots, T_{1m})$, where $T_{11}, \dots, T_{1m}$ are the subtrees of $\text{root}(T_1)$; and has right subtree $B(T_2, \dots, T_n)$



Ans: 同一種顏色表示原本為同一棵樹。



□